

Yiming Zhao

The Hong Kong University of Science and Technology (Guangzhou)

PhD Student | Guangzhou, China

Research focus: UAV vision-language navigation, radar-camera fusion, and acoustic sensing.

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Education

The Hong Kong University of Science and Technology (Guangzhou)

Guangzhou, China

PhD Student

Sep 2024 – present

IELTS: 6.5; CET-6.

Nanjing University of Aeronautics and Astronautics

Nanjing, China

Master's in Software Engineering, College of Computer Science

Sep 2020 – Apr 2023

Admitted through postgraduate recommendation; GPA: 3.71/4.00.

Henan University

Kaifeng, China

Bachelor's in Network Engineering, College of Software

Sep 2016 – Jun 2020

GPA: 3.49/4.00; top 5%.

Yuan Ze University

Taiwan

Exchange Student in Information Engineering, College of Information

Sep 2018 – Feb 2019

Publications & Manuscripts

1. **Yiming Zhao**, Tianshun Li, Jingle He, Ruonan Chai, and Xihu Zheng*. "LightVLN: Efficient Aerial Vision-and-Language Navigation with Compact Memory and History-Guided Local Aggregation." Sep 2026. Submitted to *IEEE ICRA 2027*.
2. **Yiming Zhao** and Ying Cui*. "SQRCF-UAV: Streaming Radar-Camera Fusion for 3D UAV Detection and Estimation." Apr 2026. Submitted to *IEEE ICRA 2027*; journal version submitted to *IEEE Transactions on Multimedia*.
3. Zihao Tao, **Yiming Zhao**, Hongtao Zhao, Zijun Gong, and Ying Cui*. "Sensing in Low-altitude Wireless Networks: Applications, Systems, Techniques, and Developments." Feb 2026. Accepted by *IEEE Communications Magazine*.
4. **Yiming Zhao**, Ying Cui*, Zijun Gong, and Yang Yang. "SRCF-UAV: Sparse Radar-Camera Fusion for 3D UAV Detection." Aug 2025. Accepted by *IEEE ICRA 2026*; journal version submitted to *IEEE Transactions on Multimedia*.
5. Yanchao Zhao, **Yiming Zhao***, Si Li, Hao Han, and Lei Xie. "UltraSnoop: Placement-agnostic Keystroke Snooping via Ultrasonic Sonar." Apr 2023. Published in *ACM Transactions on Internet of Things*.
6. **Yiming Zhao**, Yanchao Zhao, Huawei Tu*, Qihan Huang, Wenlai Zhao, and Wenhao Jiang. "Motion Gesture Delimiters for Smartwatch Interaction." Feb 2022. Published in *Wireless Communications and Mobile Computing*.
7. N. Zheng, Na Li*, **Yiming Zhao**, Chao Weng, and Dan Su. "Tencent Speaker Diarization System for the VoxCeleb Speaker Recognition Challenge 2021." Oct 2021. Presented at the *VoxSRC Workshop, Interspeech 2021*.

Honors & Awards

Outstanding Graduate and Advanced Research Individual (postgraduate studies)	2023
Tencent Hornbill Elite Talent Program; VoxSRC 2021 Track 4, third place (team)	2021
Network Technology Challenge, third place	2020
Excellent Student, Scholarship, Outstanding Graduate, and Outstanding Graduation Thesis (undergraduate studies)	2020

Research Experience

LightVLN: Efficient Aerial Vision-and-Language Navigation with Compact Memory and History-Guided Local Aggregation

UAV vision-language navigation

Feb 2026 – present

- Developed a policy with a 0.5B language backbone (1.276B total parameters), single-token historical memory, and history- and instruction-conditioned local aggregation; compressed 256 current-view tokens to 32, with at most 48 observation-derived tokens.
- Achieved 50.93%/36.14% success rates on OpenFly Test-Seen/Test-Unseen and 25.83% on AerialVLN-S Val-Seen.
- Validated closed-loop hardware-in-the-loop (HIL) navigation on Jetson Orin NX 16 GB, reaching 14.61 Hz model inference and 11.13 Hz end-to-end decision updates.

SRCF-UAV: Sparse Radar-Camera Fusion for 3D UAV Detection

Multimodal perception and dataset construction

Jun 2024 – Sep 2025

- Developed sparse radar-camera fusion using radar/image-assisted query initialization and distance-velocity-aware attention for real-time 3D UAV detection.
- Built a radar-camera-RTK collection platform and released a centimeter-level multimodal UAV dataset; achieved up to 91.65% mAP with approximately 17 ms latency.

UltraSnoop: Placement-agnostic Keystroke Snooping via Ultrasonic Sonar

Acoustic sensing

Oct 2020 – Oct 2022

- Used smartphone-generated chirps and keystroke reflections to estimate time differences of arrival, with two virtual coordinate systems and MFCC-based refinement.
- Achieved placement-agnostic, training-free, and keyboard-independent keystroke inference.

Industry Experience

NIO

Beijing, China

Speech Algorithms

Jun 2022 – May 2024

- Collected speech and source data, generated room impulse responses (RIRs), and constructed an in-car speech enhancement dataset.
- Improved multichannel speech separation models (RNN-BF and CLDNN) through feature, RIR, and data improvements, increasing wake-up rates by 5% over existing algorithms.
- Implemented vehicle-side acoustic echo suppression (AES) and voice activity detection (VAD), including simulation, training, and post-processing; improved TalkNet-based audiovisual lip-movement VAD. Both modules launched on NIO ET5 2.2.
- Trained and improved Qwen-based LLMs as base decoders for the team's multilingual automatic speech recognition (ASR) system.

Tencent AI Lab

Shenzhen, China

Speech Algorithm Intern

May 2021 – Nov 2021

- Improved voiceprint recognition models (ECAPA-TDNN and X-Vector) using SFace loss, mixture of experts (MoE), and dynamic pruning.
- Contributed to the team's third-place result in Track 4 of VoxSRC 2021.

Research Skills

Speech & sensing	Acoustic echo suppression, multimodal VAD, speech separation, and time-frequency analysis of speech and radar signals.
Computer vision	2D/3D detection, radar-camera fusion, UAV platforms, RTK-based dataset construction, and TI mmWave radar.
LLMs & navigation	LLM pre-training and supervised fine-tuning, multimodal models, UAV navigation, vision-language reasoning, temporal memory, and action prediction.
Frameworks	PyTorch, LoRA, and efficient model training.